

Fraud in Higher Education, Teaching Fraud Detection at
Undergraduate Level

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Introduction

This research going to present the academic corruption and fraudulent practices which become the problematic situation in last year, since the management round the world presented dedicated higher education commission to control the advanced education earners. The purpose of this research to design system for the detection and prevention framework of fraudulent behavior of teaching in the higher education at undergraduate level which spoil the future of children. Since the academic dishonesty performing by teacher during the teaching process at higher education. (de Souza-Daw, T. and Ross, R. (2021) this research proposed the academic misconduct practices and enlarges the survey over the examining authorization procedure. It is also investigated and examine that common corrupt practices in the teaching process that teacher performed during the marking of paper and assignments and teaching. Academic misconduct from teaching sides which lead to playing game over the children future. Since the blockchain auditing process which prevented changes to the system. the influence is made in the outline of detection and prevent corrupt practices in higher education by means of blockchain unchallengeable dealings. Various teaching training program incantation out the envisioned attributes of training, teaching preparation between the instructional competence and subject matter expertise required and sound character growth compulsory, (Saidin, N. and Isa, N., 2013) this research examines the teacher trainee and ethical behavior of teacher training conduct during the training process of teaching. Online survey has been conducted from 71 students enrolled in higher education studies program at university of Malaysia the resultant show that 82% cheating during the teaching process including lack of teaching practice and lack of subject knowledge over the time span of their career, this paper investigated the academic deceit between the teaching process. Since the reputation of the institution down due to the fraud behavior and cheating behavior of the higher institute.

5 EXAMPLES OF FRAUD THAT UNIVERSITIES STUDY SINCE

It included various samples of unprincipled conduct including five cases of fraud that left with multimillion-dollar hole in the higher teaching subdivision economical. (Timeshigher, 2020)

1. Lona College

In march 2011, the former vice president of finance of Lona College which situated at New York USA, based on Marie Thornton requested shamefaced to misappropriating additional \$850,000 (£525,000) through delivering university cheques for this woman personal use, by means of a university credit card for private acquisitions and creation untrue expenditures rights.

2. Vassar College

During the 2011, Arthur Fisher, the project manager at Vassar College in New York, which his wife Jennifer Fisher would sent to prison after they shaped a untrue construction company and started accusing the college for facilities that had not been did, the arrangement got \$1.9 million((£1.2 million) ended five years. (Fraud, 2022)

3. Myungshin University and Sungwha College

During the year 2011, the South Korea education ministry closed two universities down after they committed serious corruption and irregularities, which was reported Asia News, since the end of Sungwha College shadowed misappropriation by its creator, various teaching employees also involved to stolen student funds and the alternative suggest student and seek place to other universities and settle their funds.

4. The University of Montana

In the year 2010, Christine Bitterman, which employed in the residence life office at the University of Montana, begged shamefaced to misappropriating more that \$300,000 (£186,000) over historical of seven years described by Missoulain. She was theft student rent expenditures that had been made in the cash. (Fraud, 2022)

5. The University of Vermont

In 2012, Celine Berner which was former administrative assistant at University of Vermont which was involved in sent to prison after begged shamefaced deposited university cheques totaling almost \$46,000 (£28,500) hooked on her private account ended a five-year dated. Since the Burlington free press reported the university was tilted rotten by a letter from community national bank which cheque amount about \$1,425 billed to UVM Postponement - the

university's community outreach arm - that had been deposited into Ms Bernier's personal account.

Problem Statement

student take less admission over the last ten years according to research survey conducted that the research problem arises due to unethical behavior of higher education investigated during teaching process and administrative. Various intelligent student lost their degree hope due to investing their fee at wrong college. Various student affected from poor teaching behavior of teacher; the problem solves with highlight the negativity of the institutions by researching the data.

Background

according to (Petrov, G. and Temple, P., 2004) corruption at higher education some findings has been collected from Soviet Union institutional management at higher education commission. This research examines the corruption by using empirical research data. The higher education research corruption happened at Russia, Azerbaijan which arises common political framework of the teaching and administrative behavior of the management, since student does not verify the institutional reputation and investing their capital at wrong place and destroy their educational career very badly. (Rumyantseva, N.L.,2005) classification of dishonesty at higher education increases over the past five years due to poor working of education ministry to investigated the various universities. Since the paying bribes of grades, buying diplomas, including admission to universities with some fairy tale thoughts. Since the literature does not provide the sufficient data of educational corruption, so this research going to gather the research gap between the educational institutional corruption dishonesty with student and making false money to spoil their career in very bad manner. Universities corruption arises with involvement of student directly and indirectly.

1. Teaching Fraud
2. Corruption in Licensing and accreditation
3. Corruption in staff recruitment
4. Sextortion at Colleges
5. Financial management and procurement fraud
6. Political manipulation of university affair

7. Distribution operation of faculty and staff
8. Quality education drama

Research Objective

The research objective is very clear which investigated the fraud in the department of education.

Research Question

Answered at literature review section:

1. What is the planning to minimize the educational fraud over teaching, and financial?
2. How to investigate the fraud universities

Related Literature Review

(P. Raghavan and N. E. Gayar 2019) frauds are common types of activity which often used by various academic institutional but later after investigation of probe they are under investigation in the prison, various institution at Korea, Russia and Malaysia have been sealed due to poor administrative staff and management. Additionally, they bypass security check leading the loss of millions of dollars of student community over the world, somehow student does not investigate their educational background history and are not able to investigated their overall outcome. Data mining technique focusing on tracing fraudulent transactions. Hence detecting unusual activities this research present data mining method to solve the complexity of fraud education. Such as KNN, SVM, Decision tree algorithm used to prove the dataset. Hence the very few literatures appear to investigate the fraud educational institutes. (Lincke, S. and Green, D., 2012.) teaching case studies appear in this research, since the people are becoming more creative to investigate the classic fraud schemes of educational institutes over teaching and administration scheme. Financial scams happened with asking student to pay them for rent and extra mileage KM allowance detected from their fee. Inappropriate assignment checking which leads towards the fraud and scam various dataset view in investigated the Malaysian scam universities. Since the information security officers help in investigating the teaching fraud the first case studies designed to probe the undergraduate student scam at department of computer

science and information technology. The information security officers investigated the scholar offices by mean of asking some information and plan some educational in view of curriculum. The design of studies scheme has been investigated to social engineering attack by sending spam emails to students since the security design response department investigated the incident response in which various administrative staff involve to scam the student and destroy their future. (Aziz, R.H.A. and Jusoff, K., 2009.) effective poster teaching strategies teaching towards risk in studying fraud. This research presents another method in teaching and learning at higher institutes its was very difficult problem to detect the teaching fraud since the very few investigations are highlighted in the research. the lecture delivering policy only rely on visual aid hence not providing the solicited example by hand writing, this research suggest some novel teaching methods by using the visual aid and visual poster aid in which student can takes interest to learn new advance information technology tool, but still its not complete solution to facilitate the student because they invest fee in their career development. (Lyles, and M.A., 2013) teaching medical student about fraud and medical mortality hence the program integrity designed by mean of comprehensive strategies to detect fraud, abuse, errors and waste in the US. Since the healthcare system in US eliminating inappropriate payment by mean of fraud detection to detect the fraud officers by administrative surprise visit at center. Hence the institute of medicine proved in 2012, the waste fraud annually losses of 765\$ billion US dollar including improper disbursement of unnecessary services of administrative team, and the administrative team delivering unnecessary services to the team by facilitating them without any reason and getting price.

[Sextortion in Education Sector Tanzania](#)

(Yusuph, K., 2016) this research investigated sextortion at educational institutes at Tanzania in response to criminal legal system. since the criminal justice at Tanzania detected various sexual activities by engaging teacher between the students during off schedule of classes at summer. This result method focusing on results by available literature and social media reports to address important issues such as sexual corruption at education sector by mean of degree issuance. Hence the legal administrative department taken action against such criminal institutes and make justification against the sexual activities to capture CCTV recording beside this newspaper are

biased and unauthentic for spreading some fake news about the student sex. Hence the sexual harassment against women which was common world problem. Hence the research investigated the sex distortion from primary level to higher education level not in Tanzania but other countries are also facing such types of crime. This research concluded that number of sextortion has been made and file at court and judgement of justification has been made during the investigation of prison of university management.

Corruption in School and Universities at Kenya

Corruption is key issues at Kenya (Taaliu, S.T., 2017) hence in all sector such as health, administrative authority and education sector. Hence the alleged corruption has been during recruitment process such as promotion, deployment and transfer of teacher. Since the primary school and secondary school facing corruption during promotion of teacher and increment of salary hence the common problem detected such as ethnicity and face color and tribalism in employment promotion and recruitment of students as well.

Manifestation of Corruption at Higher Education and the role of university

Corruption has been found in (Seniwoliba, J.A. and Boahene, B.E., 2015) variety of form and different dimensions since the qualitative research method has been made in investigating the educational corruption to elicit the result. This research present the policy document and made transparency international report by probing the investigation of corruption, hence the universities administration involved in making corruption in critical perspective at public sector of GHANA .since the review of corruption based on idealism and rational choice, in the research major types of corruption attribute founds in research such as admission, procurement, leadership recruitment process and academic dishonesty by mean of cheating , leaking of examination paper plagiarism, favourism etc. at the end the research recommends that the government should take initiative by investigation the aged administrative staff at educational level and taking extra salary by mean of their old age group and taking special attention at Ghana moral values.

Corruption at Universities Paths to integrity at higher education

(Kirya, M., 2019) since the many developing countries ruled by army, so the common problem arises due to insufficient administrative staff which does not have sufficient knowledge about

education and such meaning of teaching, they only considered visual aid methods, power point slides methods to teach and scam the students, hence it proved that the power point slide based method is not good for teaching because it does not provide the practical skills to the student community in which they take interest practically to solve the common programming error. In department of computer science making insufficient practical knowledge of programming to solve the complexity of algorithm so it was not considered that the practical skills of student have been lack due to slide based learning not so good. Hence the abuse of private gain investing funds at wrong place without any reason to solve the complexity of institutions.

Public University at Kenya Ethical issues in Recruitment selection and employee performance

According to this research (Sarah, M., Sang, A. and Ngure, S.W., 2018.) ethics plays essential role in the process of recruitment selection and teaching. Hence the teaching staff properly hired and managed by investigating their educational background and investigation their social moral values and cultural background. Hence the ethical consideration ensue that the candidate is review according to their educational merit. The main objectives of this research to examine the recruitment and selection process by mean of educational committee which takes place under investigation committee to hire the teacher. This research used stratified random sampling including descriptive research design based on statistical method to proof the dataset of Kenya public sector universities.

Corruption in Higher Education

(Heyneman, S.P., 2020) corruption in higher education are very common at higher studies scheme level due to insufficient staff team which have very few knowledges about the staff recruitment and management. Hence the academic dishonesty and academic misconduct has been common practice of the higher education institutes. Since the dishonesty of the academic staff to asking to pay for the exam fee extra without any reason and failure the student in their assignments without any reason. Since the transparency international working at worldwide to minimize the corruption at various level when the institution perceived they are corrupt they damage the society already, hence the corruption at university was happened the involvement of student with and without.

Research Methods

Dataset Information & Interpretation

Dataset collected from government of South Australia published data at 2019, and updated in 2021, published agency is education ministry. The title of dataset is “Fraud detection in the department of Education. Dataset updated frequency annual which added and remove attribute after some period of time. (Education, 2019).

Data Collection Method

Primary Dataset

This data collected from Ministry of Education South Australia published at 2019 and updated in 2021.

Secondary Dataset

Secondary research data collected from Google Scholars, IEEE conference paper, IEEE Xplore Site, ACM digital library including Science Direct.

Research Positivism

This research positivism used the statistical methods to proof the dataset which controlled experiment based on the machine learning pattern. The research positivism used the true facts and knowledge to investigate the problem by appropriate way of research practices.

Quantitative Method

This research presents the quantitative research method to explore the dataset which downloaded from the South Australia site, it uses the deductive approach to find the reasoning, examined the research question, the dataset clearly communicated with numerical facts and proof in data mining algorithm

Data Mining & Statistics

1. Multivariate data analysis
2. Bivariate data analysis

3. Statistical analysis

Exploratory Data Analysis

Exploratory data analysis useful method to view the statistical facts of the dataset and view the visualization method to predict and investigate the attribute of research problem.

Attribute of dataset

1. Instances of fraud detected in the Department for Education
2. Category/nature of fraud
3. Potential theft
4. Potential misappropriation of funds
5. Inappropriate financial processes
6. Falsification of attendance data to claim fees
7. Fraudulent claiming of kilometers
8. Fraudulent behavior or misrepresentation by an employee.

Dataset information 2013 to 2020.

Data preprocessing in python

```
In [37]: fraud_data= pd.read_csv('annual-report-open-data_fraud_2013_2020.csv')
```

```
In [38]: fraud_data
```

```
Out[38]:
```

	Instances of fraud detected in the Department for Education	year2013	year2014	year2015	year2016	year2017	year2018	year2019	year2020
0	Category/nature of fraud	2013.0	2014.0	2015.0	2016	2017.0	2018.0	2019.0	2020.0
1	Potential theft	3.0	5.0	6.0	2	0.0	0.0	3.0	1.0
2	Potential misappropriation of funds	3.0	2.0	2.0	1	1.0	1.0	1.0	0.0
3	Inappropriate financial processes	1.0	0.0	0.0	0	2.0	0.0	1.0	4.0
4	Falsification of attendance data to claim fees	2.0	2.0	2.0	2*	0.0	0.0	1.0	1.0
5	Fraudulent claiming of kilometres	0.0	0.0	0.0	1*	0.0	0.0	0.0	0.0
6	Fraudulent behaviour or misrepresentation by a...	0.0	0.0	0.0	0	2.0	7.0	2.0	3.0

Dataset value counts

```
In [41]: fraud_data['year2014'].value_counts()
```

```
Out[41]: 0.0      3
          2.0      2
          2014.0    1
          5.0      1
          9.0      1
          Name: year2014, dtype: int64
```

Data Cleaning.

```
In [44]: #data cleaning
```

```
In [45]: fraud=fraud_data.replace('?',np.nan)
```

```
In [46]: fraud.head()
```

```
Out[46]:
```

	Instances of fraud detected in the Department for Education	year2013	year2014	year2015	year2016	year2017	year2018	year2019	year2020
0	Category/nature of fraud	2013.0	2014.0	2015.0	2016	2017.0	2018.0	2019.0	2020.0
1	Potential theft	3.0	5.0	6.0	2	0.0	0.0	3.0	1.0
2	Potential misappropriation of funds	3.0	2.0	2.0	1	1.0	1.0	1.0	0.0
3	Inappropriate financial processes	1.0	0.0	0.0	0	2.0	0.0	1.0	4.0
4	Falsification of attendance data to claim fees	2.0	2.0	2.0	2*	0.0	0.0	1.0	1.0

Descriptive Statistics

```
In [53]: #Statistics
```

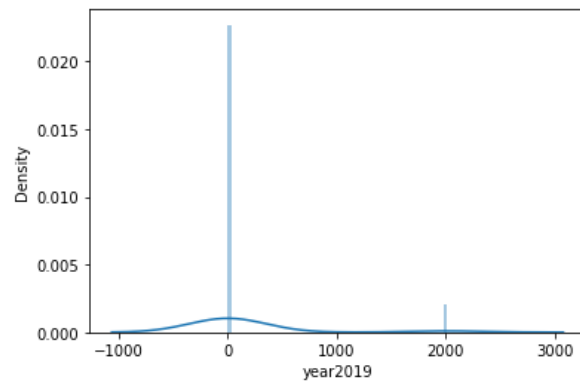
```
In [54]: fraud.describe().T
```

```
Out[54]:
```

	count	mean	std	min	25%	50%	75%	max
year2013	12.0	170.083333	580.372758	0.0	1.75	2.5	3.00	2013.0
year2014	12.0	170.000000	580.714763	0.0	1.50	2.0	2.75	2014.0
year2015	12.0	170.250000	580.952529	0.0	1.50	2.0	3.00	2015.0
year2017	12.0	169.416667	581.839397	0.0	0.75	1.5	2.00	2017.0
year2018	12.0	169.666667	582.080724	0.0	0.00	0.5	2.50	2018.0
year2019	12.0	170.083333	582.261259	0.0	1.00	1.5	2.25	2019.0
year2020	12.0	170.500000	582.446330	0.0	1.00	2.0	3.25	2020.0

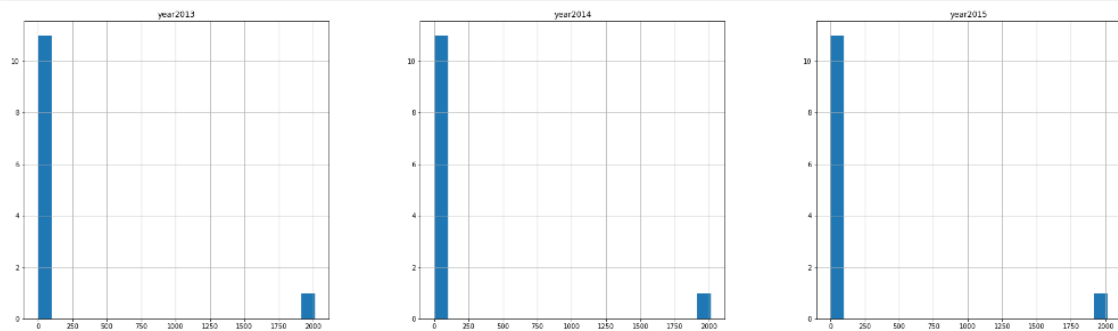
Bivariate Analysis

```
Out[56]: <AxesSubplot:xlabel='year2019', ylabel='Density'>
```

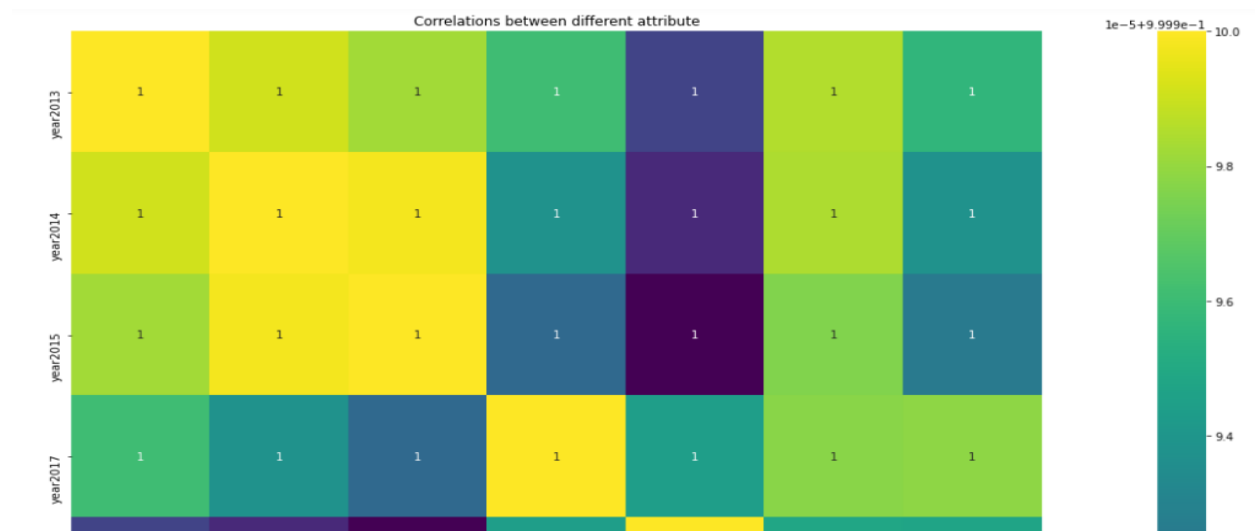


Multivariate analysis

```
In [58]: fraud.hist(bins=20,figsize=(30,30), layout=(3,3));
```



Correlation between attributes



Results & Discussion

(Mattar, M.Y., 2021) combining academic corruption which was taking serious note to identify the common corrupt element at education level, most of the student complaint by survey that teacher does not have sufficient lecture knowledge and command over the subject to facilitate the student for teaching and practicing them. Student complain about the visual aid and power point slide is not good tool for teaching the programming lecture. Hence, it's not provided the assignment solution with solid reason of practical and sound background of knowledge driven approach. Since the academic integrity identified by international accreditation center in the model of Qatar university. Since the academic could gain good feedback from the student community when they delivered quality education to the student and delivered quality standard practical skills with sufficient proof. Academic integrity defined at cheating and plagiarism policy this research examines the various academic year corruption chart by making the south Australia education government. Compliance the standard of human rights, labor rights and education rights which was commonly identified by the institutional to deliver the esteem level of lecture towards the student community which takes part of education interest and making their better career. Since the academic management should be able to design the academic standard chart based on the recruitment and selection of teaching and selection of good student as well.

Hence the Qatar international university adopted the international standard of educational based on the facilitating student criteria and recruitment policy of the teacher as well. The corruption chances is 0% due to the transparency level is very high.

(Denisova-Schmidt, E., Huber, M., Leontyeva, E. and Solovyeva, A., 2021) computing experimental evidence with machine learning to assess the anti-corruption educational system at Russian university since the academic integrity run in short term to facilitating them to the student with academic standard goal. Since the 2000 survey samples collected random to assess the teaching facilities at academic standard or not. The negative point highlighted to improve the educational institutes with some solid reason by replacing the old academic staff with young teaching and including good educational merit including research skills. Student might able to react at some points to provides the solid reason of the educational planning.

Conclusion

Hence the attributes of dataset taken from the research as follows

1. Instances of fraud detected in the Department for Education
2. Category/nature of fraud
3. Potential theft
4. Potential misappropriation of funds
5. Inappropriate financial processes
6. Falsification of attendance data to claim fees
7. Fraudulent claiming of kilometers
8. Fraudulent behavior or misrepresentation by an employee.

Analyzed them in python programming and make suggestion by exploratory data analysis that there is need to improve the education system at Nigeria, Kenya, Uganda, South Korea, Russia and Malaysia. It was found in research literature that various corruption has been happened at these mentioned countries because of the staff is not upto the educational merit standard so it is suggested that there is need to improve the educational teaching staff to facilitating the student in education history. Since the educational staff does not provide the quality education at mentioned countries and the corruption was proved in different level of dimension and prevent by investigating and all management under prison. So it is recommended that by survey the literature that annual meeting conducting to review the teaching quality by investigating them feedback from the student community.

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